

# Sampling contacts from sequence

A figure-first writeup on existing predictors, oracle contacts, MarinFold and useful sampling diversity.

Fixed model: exp277 step 266344, 1.47B parameters, 248.584B raw training tokens. Natural proteins lead: 97 validation + 217 publication test. Designs remain separate. AF2/AF3/Boltz-2 comparators use shared archived MSAs and no templates, with confidence selection.

Predictor generation → cached analysis tables → fast static and interactive rendering. Every numerical figure element maps back to a source CSV row.

# What the contact and confidence analyses show

AF2 / AF3 / Boltz-2 GDT-TS: 0.854 / 0.860 / 0.861 on 305 natural proteins. At MSA depth <10: 0.196 / 0.332 / 0.239, with only five proteins.

MarinFold contact R-precision: 0.561 overall; 0.345 at MSA depth <10 versus 0.617 at  $\geq 1000$ . Only five natural proteins occupy the shallowest tier.

Oracle pTM ranks 1, 5, 1, 1, 1 out of 101 maps for the five low-depth proteins. Highest pTM selects each map's Helico sample; no ipTM or clash penalty. 100 ESMFold2 maps/protein, a shared eligible-pair mask, three Helico samples/map. Original ESMFold2 median TM-score: 0.848, 0.860, 0.881, 0.649, 0.839. pTM-source TM-score Spearman: 0.127, 0.083, 0.254, 0.288, 0.613. Protein order: 8ii8\_A, 8oxk\_A, 8qoh\_A, 8ux2\_A, 8wrx\_A. Only five biological examples.

Top-L contacts raise Helico GDT-TS from 0.150 to 0.504 on 305 matched natural proteins. The paired gain is 0.354 [0.323, 0.387]. All cuts were fixed before test inference.

# Can more AlphaFold3 sampling recover accurate folds?

5,000 predictions: 1,000 independent full runs for each of five natural test proteins. One diffusion sample per seed, ten recycles, fixed archived MSA, no templates.

8oxk\_A: best TM 0.516 in the original 25 → 0.818 in 100 fresh runs → 0.947 in 1,000. pTM selects the 0.947 structure (seed 10470, draw 471). 20 / 1,000 predictions reach  $TM \geq 0.8$ ; the first is draw 85. At budget 100, pTM selected a different structure with TM 0.643.

8qoh\_A / 8ux2\_A: no  $TM \geq 0.8$  in 1,000; best TM 0.502 / 0.635. 8ii8\_A / 8wrx\_A: already accurate; best TM 0.881 / 0.936.

Per-protein plots separate oracle best-of-N from pTM-selected TM. The first 100 runs are unchanged; original baseline scores reproduce. Five biological examples, fixed generation settings, no model tuning.

# Useful diversity is the open question

314 proteins  $\times$  100 samples: consensus R-precision 0.561; oracle best individual map 0.528. Paired difference  $-0.032$ , 95% bootstrap interval  $[-0.038, -0.025]$ .

All 100 maps are distinct for every protein; mean Jaccard 0.277, true-contact union recall 0.943. This is not an absence of contact-set diversity, and it does not directly measure distinct folds.

Next: better whole-map candidates, inference-time search and post-training.

# Low benchmark MSA depth versus ESMC training sources

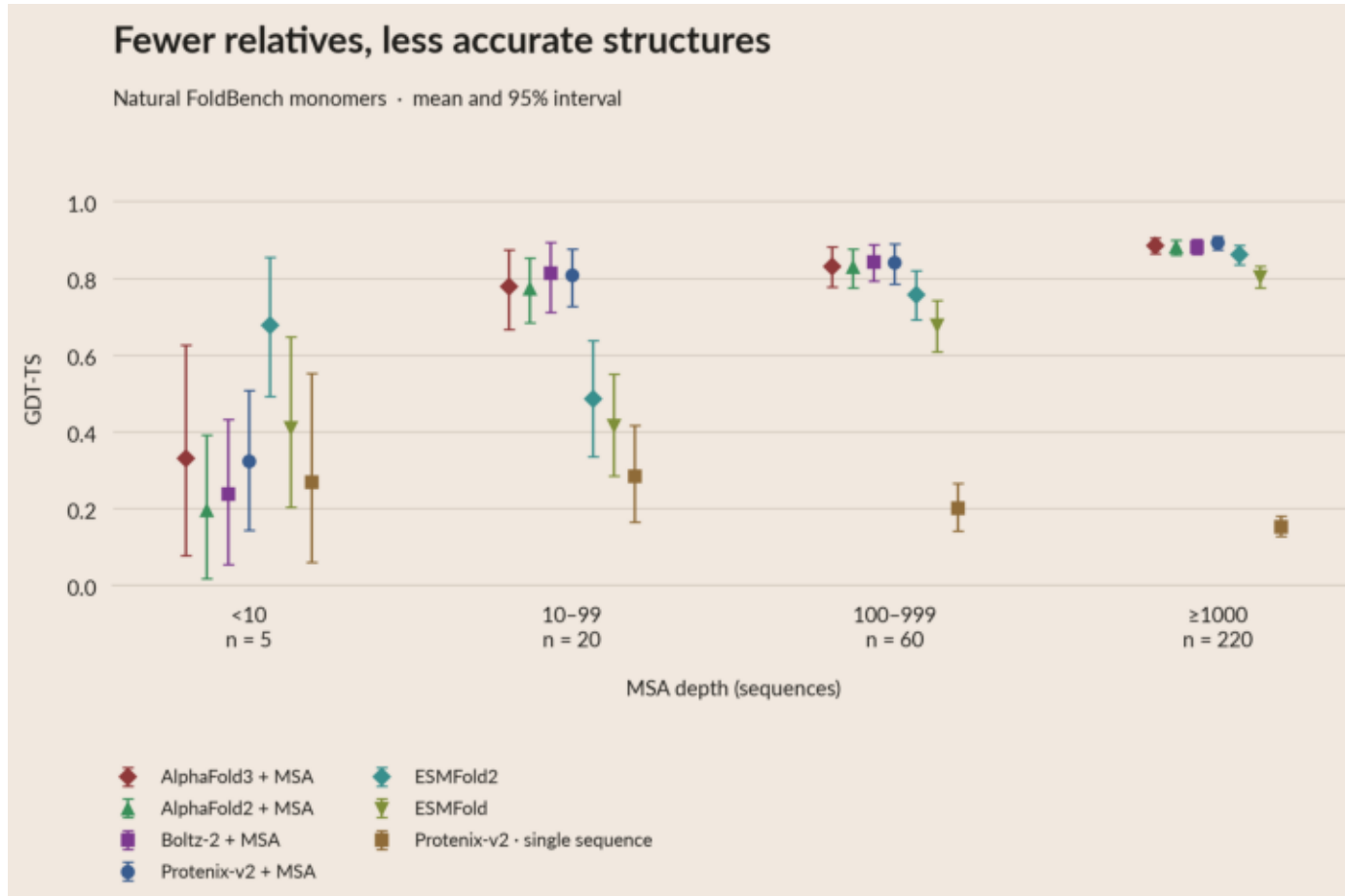
ESMC documents UniRef 2023\_02, MGnify 2023\_02 and JGI (July 2023), not Logan. Low retrieved alignment depth does not establish novelty to the language model.

New historical UniRef search, with MSA filtering disabled: Significant hits covering at least half the query: 1, 4, 1, 15, 1. Protein order: 8ii8\_A, 8oxk\_A, 8qoh\_A, 8ux2\_A, 8wrx\_A. 14 of the 15 well-covered 8ux2\_A hits are 97.9–100% identical to the query. Counts exclude the added query; they are not Neff or final training-cluster counts.

Online MGnify30 hits count heavily clustered representatives, not training-source depth. ESMC uses 70%-identity clustering; historical MGnify90 is the closer public search target. Full historical MGnify and JGI are not yet searched. Complete training depth is unknown. See TRAINING\_SOURCE\_SEARCH.md and data/training\_source\_depths.csv for details. Offline reduction: `uv run python generation/search_training_sources.py prepare`

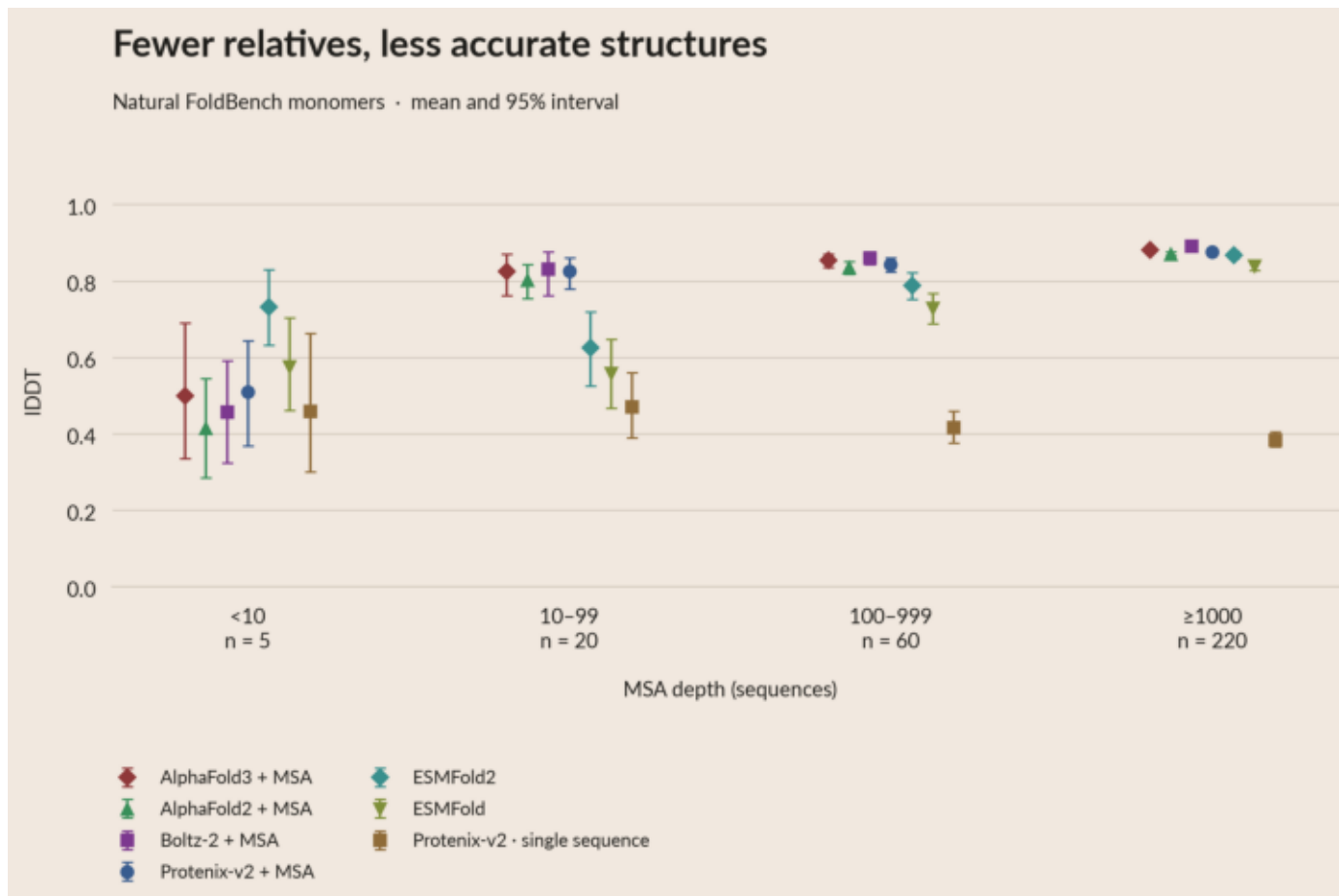
# 01\_predictors.png

Natural FoldBench monomers; the same 305 proteins in every structural arm. Points are protein means; bars are 95% protein-bootstrap intervals. MSA depth counts sequences, including the query, in the alignment used by Protenix-v2 + MSA. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



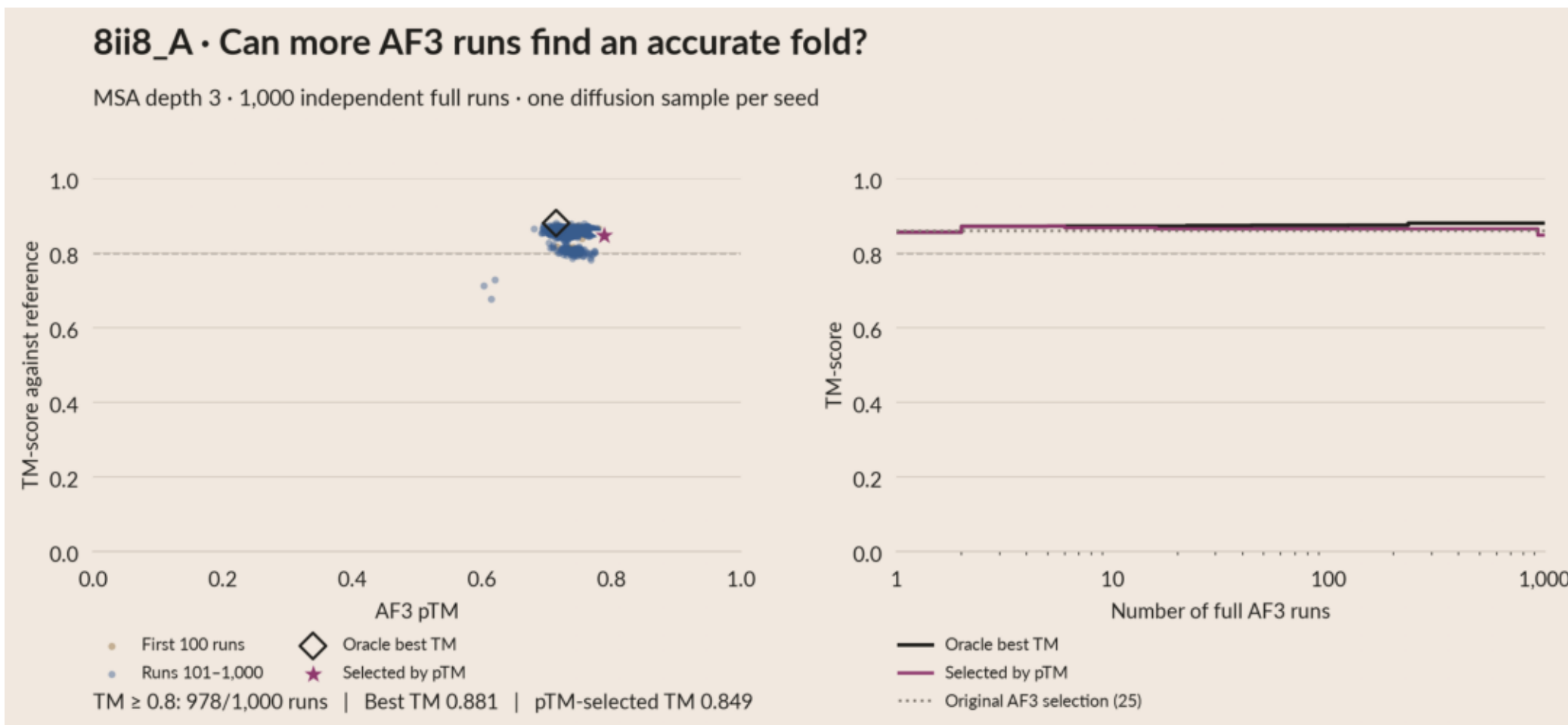
# 01\_predictors\_lddt.png

Natural FoldBench monomers; the same 305 proteins in every structural arm. Points are protein means; bars are 95% protein-bootstrap intervals. MSA depth counts sequences, including the query, in the alignment used by Protenix-v2 + MSA. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



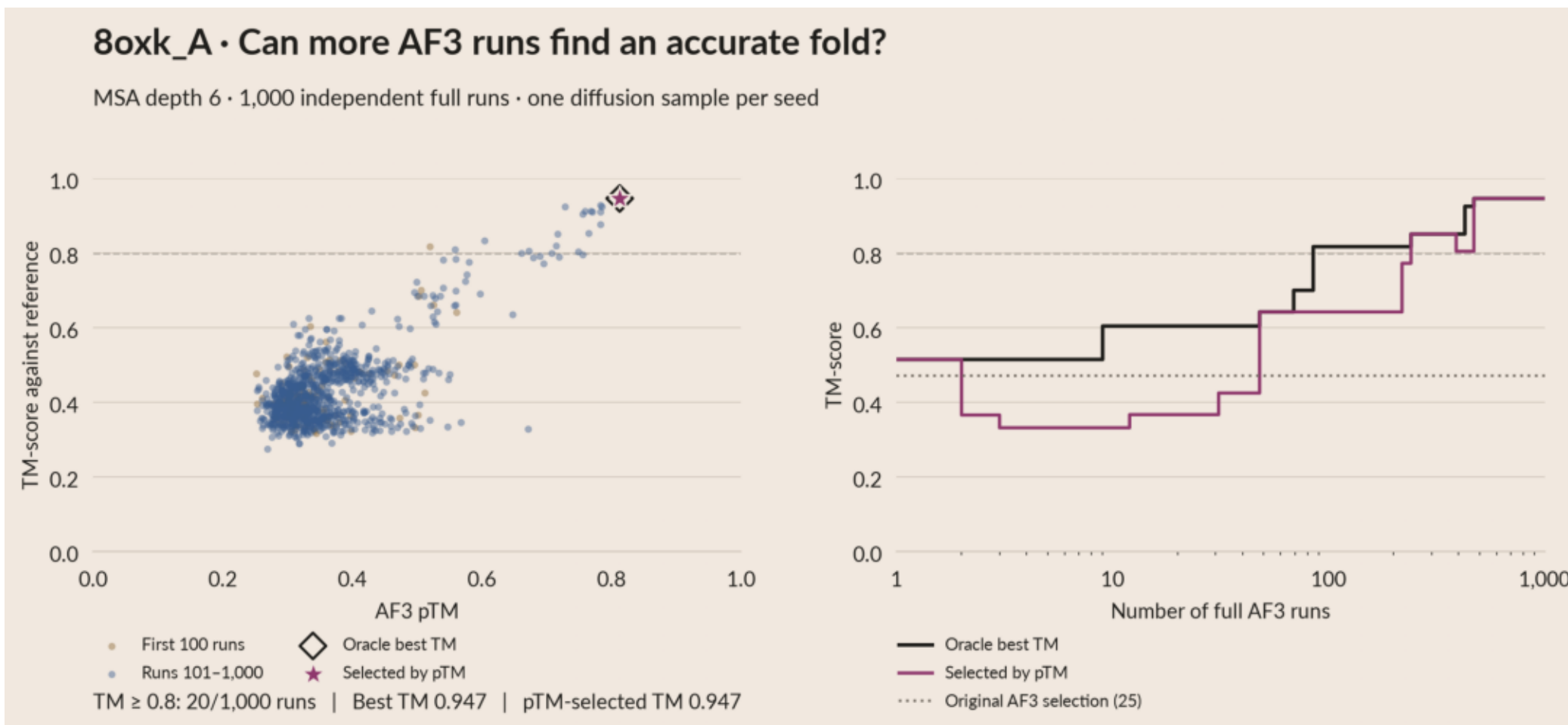
## 01b\_af3\_sampling\_8ii8\_A.png

8ii8\_A: each point is one independent AF3 trunk + diffusion run, seeds starting at 10000, ten recycles, fixed archived MSA and no templates. Full-precision pTM selects without using truth. Oracle best TM uses truth only for this diagnostic. Dashed line: prespecified TM  $\geq 0.8$ . Curves use the ascending-seed prefix; original dotted baseline selected among five seeds x five samples by official ranking\_score. Data: af3\_sampling\_samples.csv and af3\_sampling\_curves.csv; original baseline: af3\_sampling\_original.csv. All are in data/; preprocessing: prepare\_af3\_sampling.py.



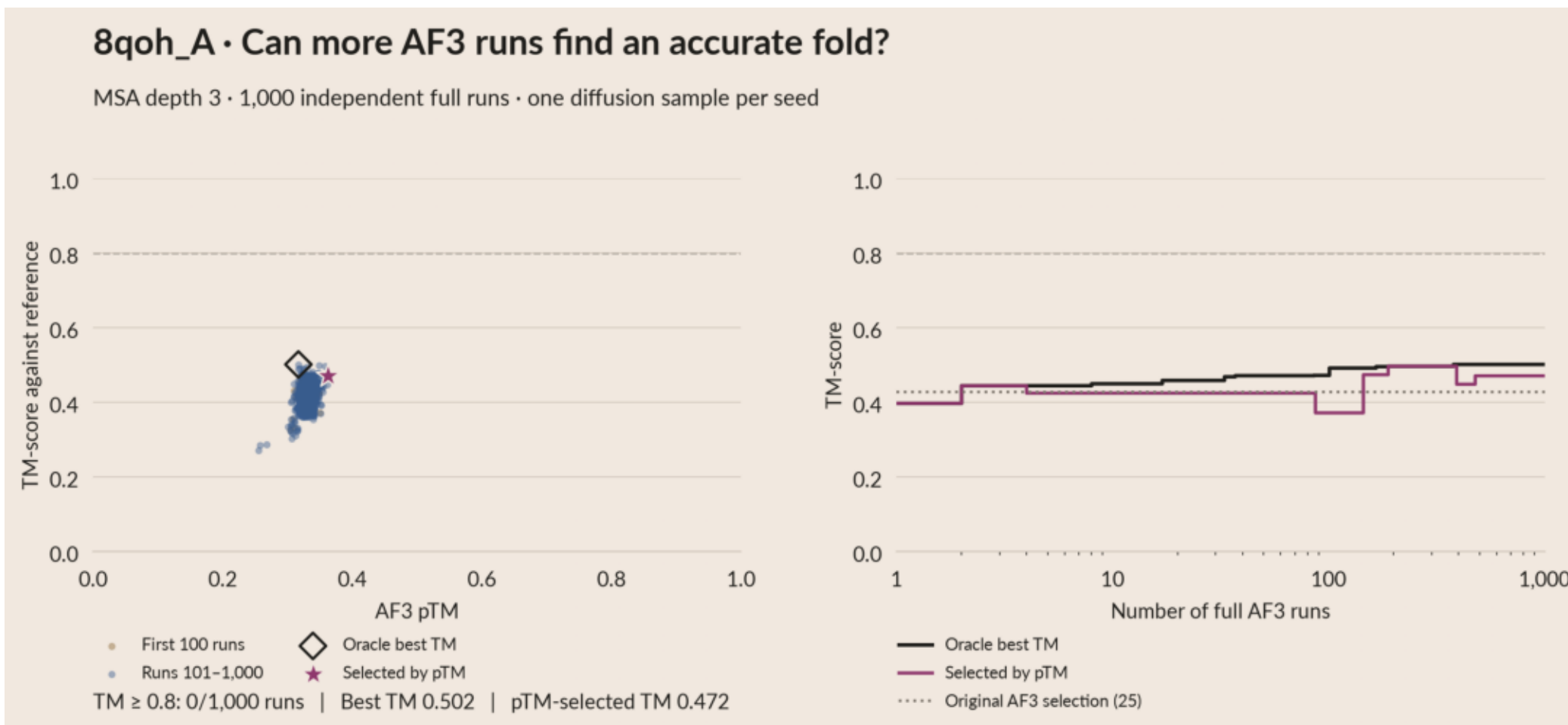
## 01b\_af3\_sampling\_8oxk\_A.png

8oxk\_A: each point is one independent AF3 trunk + diffusion run, seeds starting at 10000, ten recycles, fixed archived MSA and no templates. Full-precision pTM selects without using truth. Oracle best TM uses truth only for this diagnostic. Dashed line: prespecified TM  $\geq 0.8$ . Curves use the ascending-seed prefix; original dotted baseline selected among five seeds x five samples by official ranking\_score. Data: af3\_sampling\_samples.csv and af3\_sampling\_curves.csv; original baseline: af3\_sampling\_original.csv. All are in data/; preprocessing: prepare\_af3\_sampling.py.



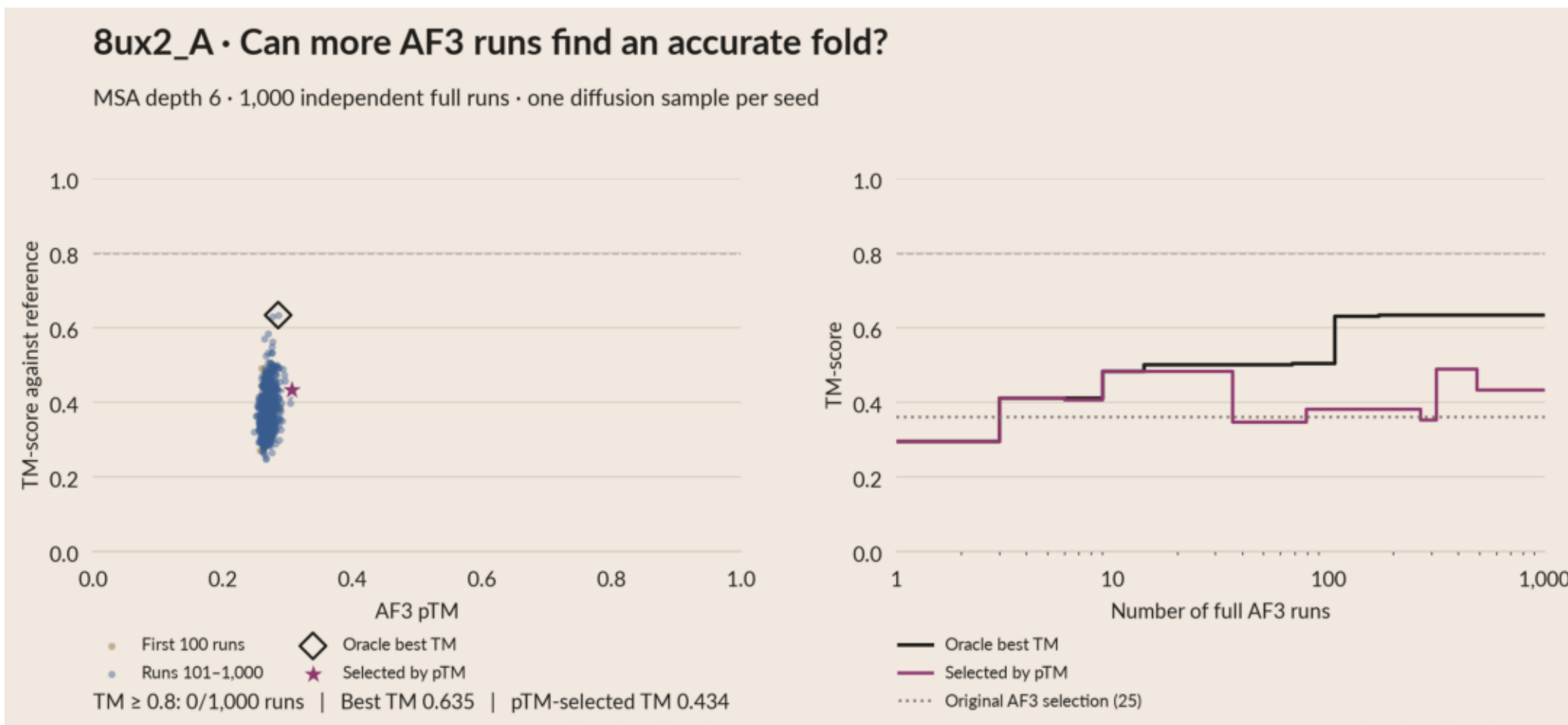
## 01b\_af3\_sampling\_8qoh\_A.png

8qoh\_A: each point is one independent AF3 trunk + diffusion run, seeds starting at 10000, ten recycles, fixed archived MSA and no templates. Full-precision pTM selects without using truth. Oracle best TM uses truth only for this diagnostic. Dashed line: prespecified TM  $\geq 0.8$ . Curves use the ascending-seed prefix; original dotted baseline selected among five seeds x five samples by official ranking\_score. Data: af3\_sampling\_samples.csv and af3\_sampling\_curves.csv; original baseline: af3\_sampling\_original.csv. All are in data/; preprocessing: prepare\_af3\_sampling.py.



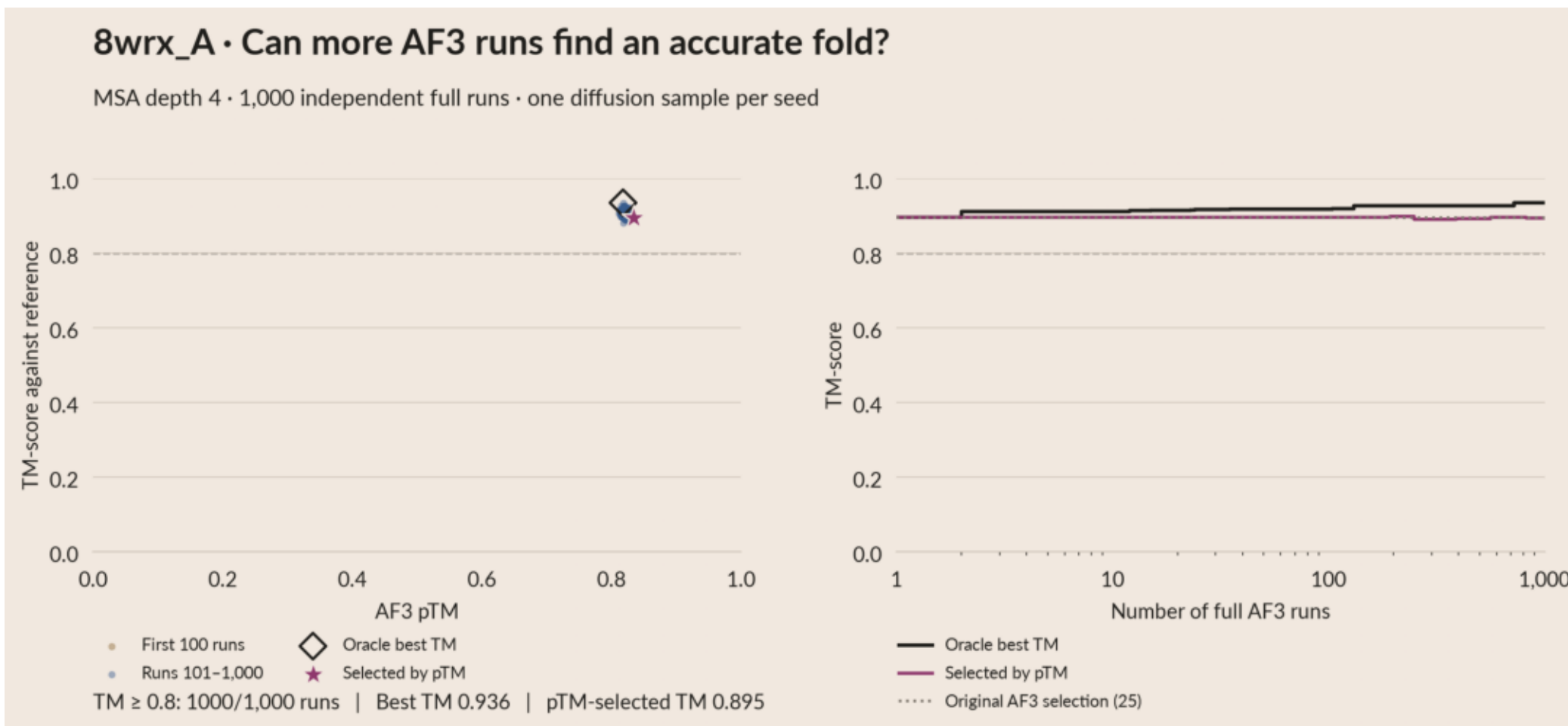
## 01b\_af3\_sampling\_8ux2\_A.png

8ux2\_A: each point is one independent AF3 trunk + diffusion run, seeds starting at 10000, ten recycles, fixed archived MSA and no templates. Full-precision pTM selects without using truth. Oracle best TM uses truth only for this diagnostic. Dashed line: prespecified TM  $\geq 0.8$ . Curves use the ascending-seed prefix; original dotted baseline selected among five seeds x five samples by official ranking\_score. Data: af3\_sampling\_samples.csv and af3\_sampling\_curves.csv; original baseline: af3\_sampling\_original.csv. All are in data/; preprocessing: prepare\_af3\_sampling.py.



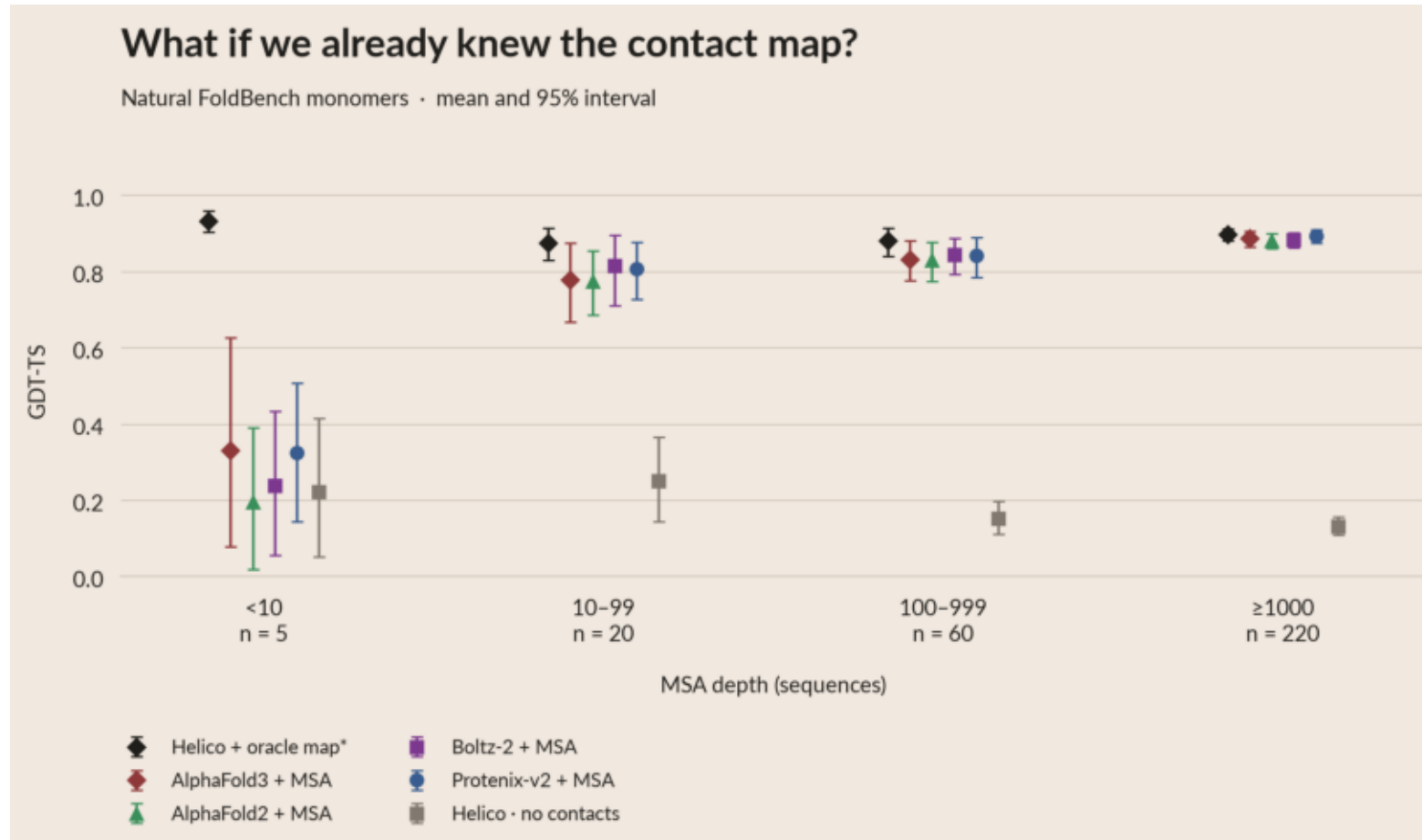
## 01b\_af3\_sampling\_8wrx\_A.png

8wrx\_A: each point is one independent AF3 trunk + diffusion run, seeds starting at 10000, ten recycles, fixed archived MSA and no templates. Full-precision pTM selects without using truth. Oracle best TM uses truth only for this diagnostic. Dashed line: prespecified TM  $\geq 0.8$ . Curves use the ascending-seed prefix; original dotted baseline selected among five seeds x five samples by official ranking\_score. Data: af3\_sampling\_samples.csv and af3\_sampling\_curves.csv; original baseline: af3\_sampling\_original.csv. All are in data/; preprocessing: prepare\_af3\_sampling.py.



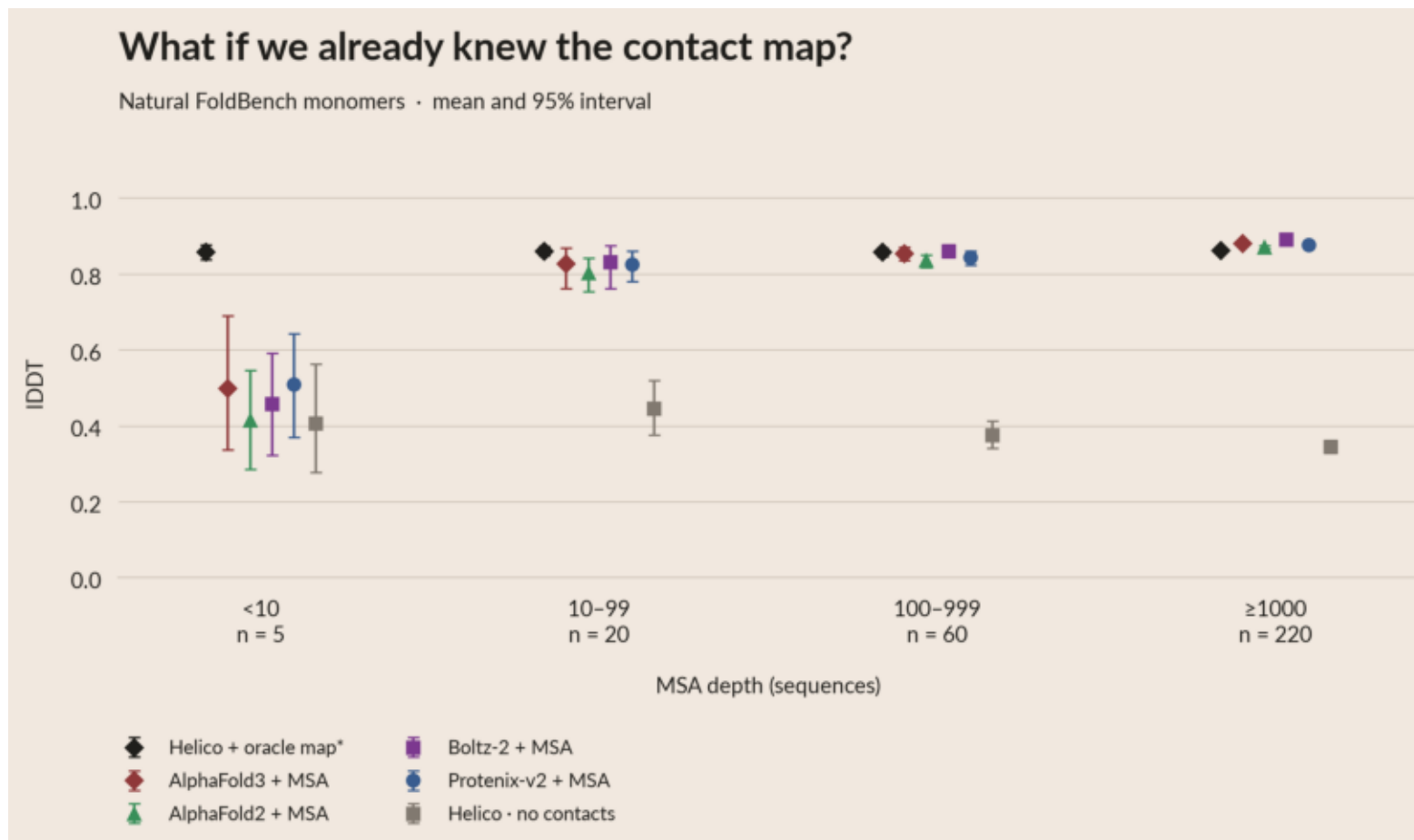
## 02\_oracle.png

\*Oracle = ground-truth contacts AND non-contacts supplied to Helico without an MSA. This is an information upper bound, not a competing predictor or a matched-count true-contact experiment. Same 305 natural proteins as Figure 1. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



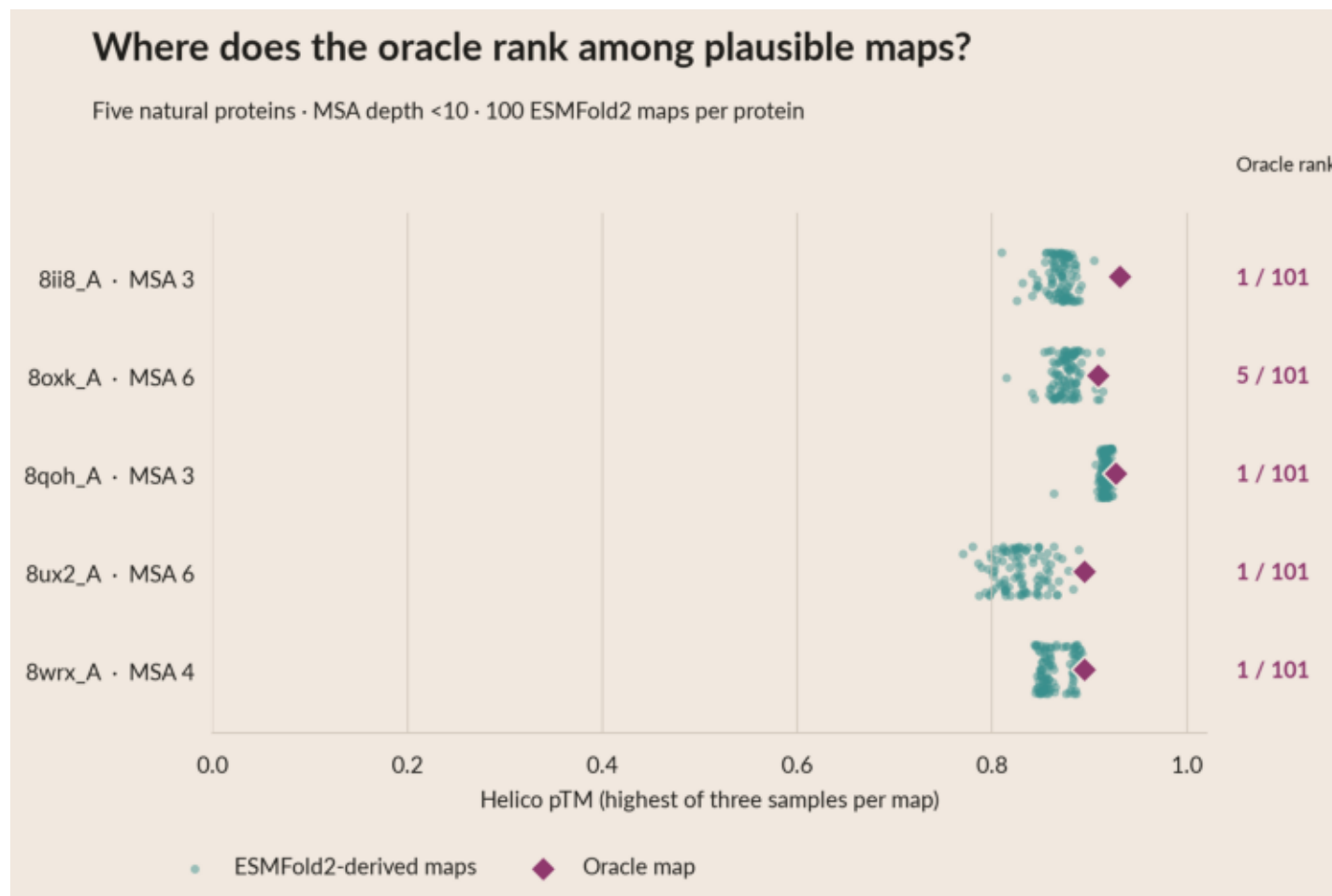
## 02\_oracle\_lddt.png

\*Oracle = ground-truth contacts AND non-contacts supplied to Helico without an MSA. This is an information upper bound, not a competing predictor or a matched-count true-contact experiment. Same 305 natural proteins as Figure 1. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



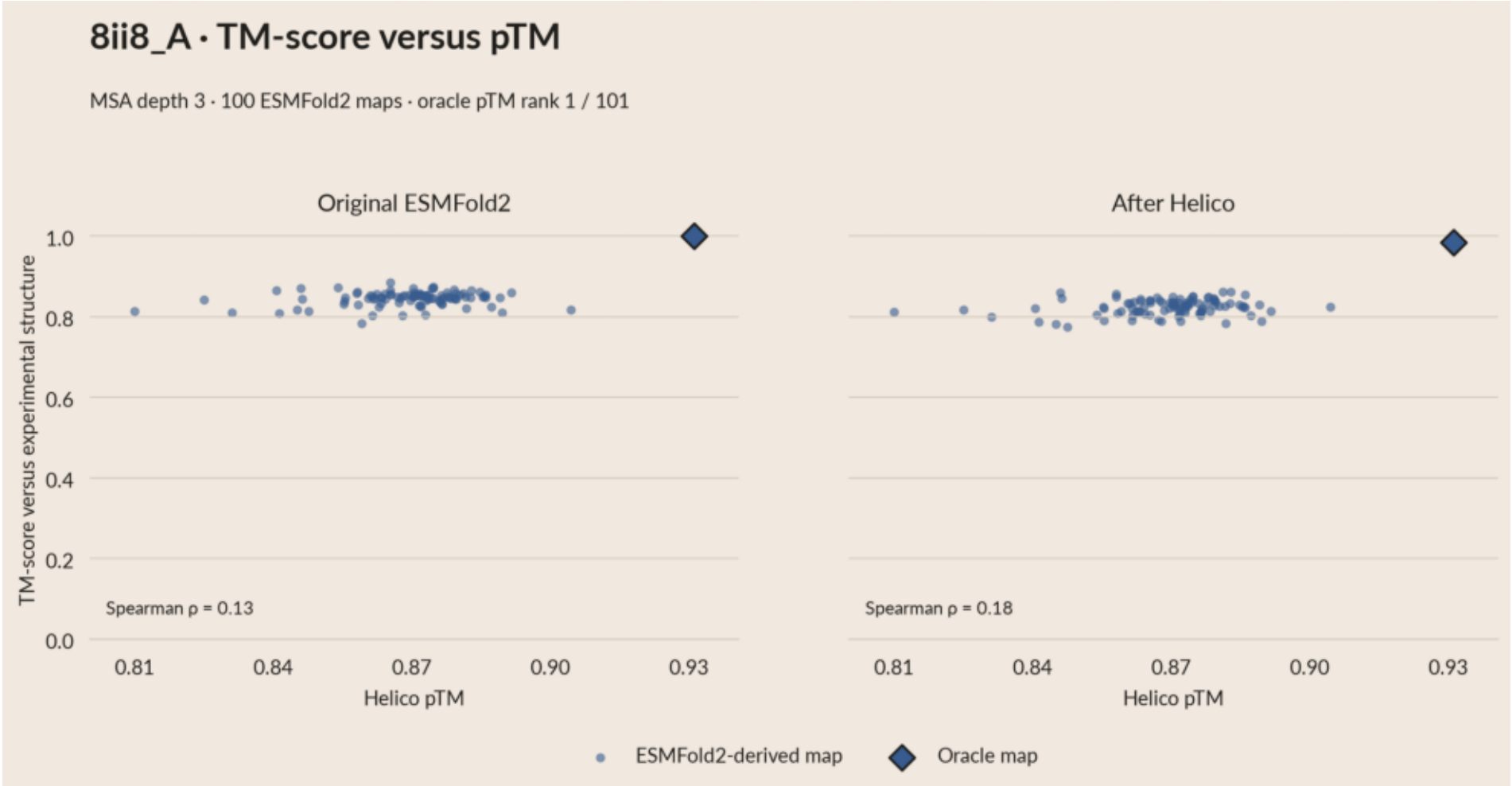
## 02b\_confidence.png

All five natural FoldBench proteins with MSA depth <10 (test split). Each dot is one of 100 single-sequence ESMFold2 predictions, converted to a full contact/non-contact map and folded by Helico; diamonds use the ground-truth oracle map. Every map has the same eligible-pair mask and three Helico diffusion samples. Highest pTM selects a sample per map; that pTM also ranks the 101 maps. No ipTM term, clash penalty or map filtering is used. Contact counts may vary. Rank 1 is highest pTM; ties are shown as rank intervals. Vertical jitter only separates points. These 500 decoys represent five proteins.



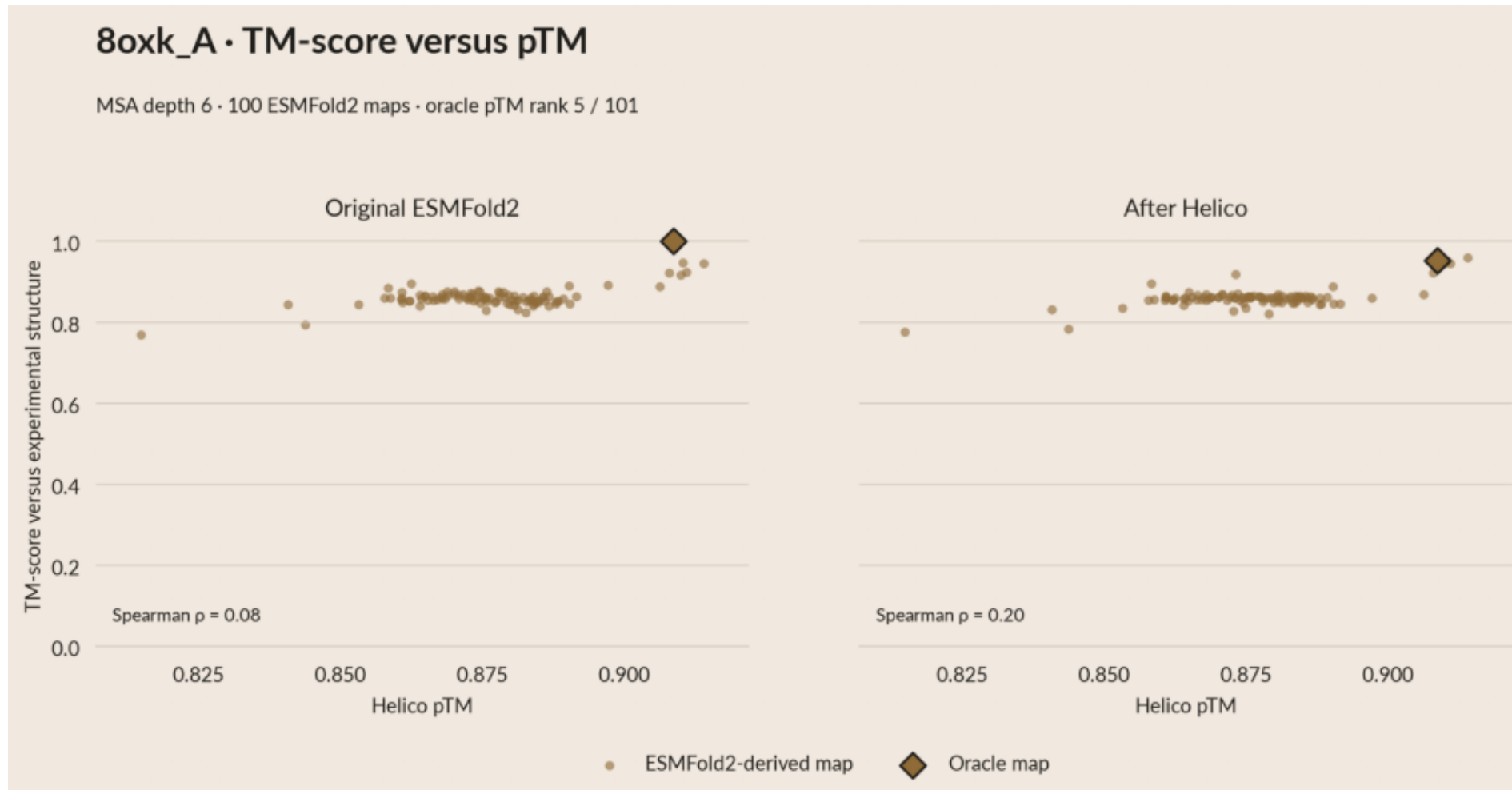
# 02c\_accuracy\_confidence\_protein\_8ii8\_A.png

Highest pTM selects one of three Helico samples for every map; pTM also ranks the 101 maps. Left: original ESMFold2 TM-score (oracle reference = 1). Right: selected Helico reconstruction TM-score. Diamonds identify oracle contacts; correlations use only the 100 ESMFold2 maps. TM-score uses matched protein CA atoms; pTM retains all Helico input tokens. No ipTM or clash penalty.



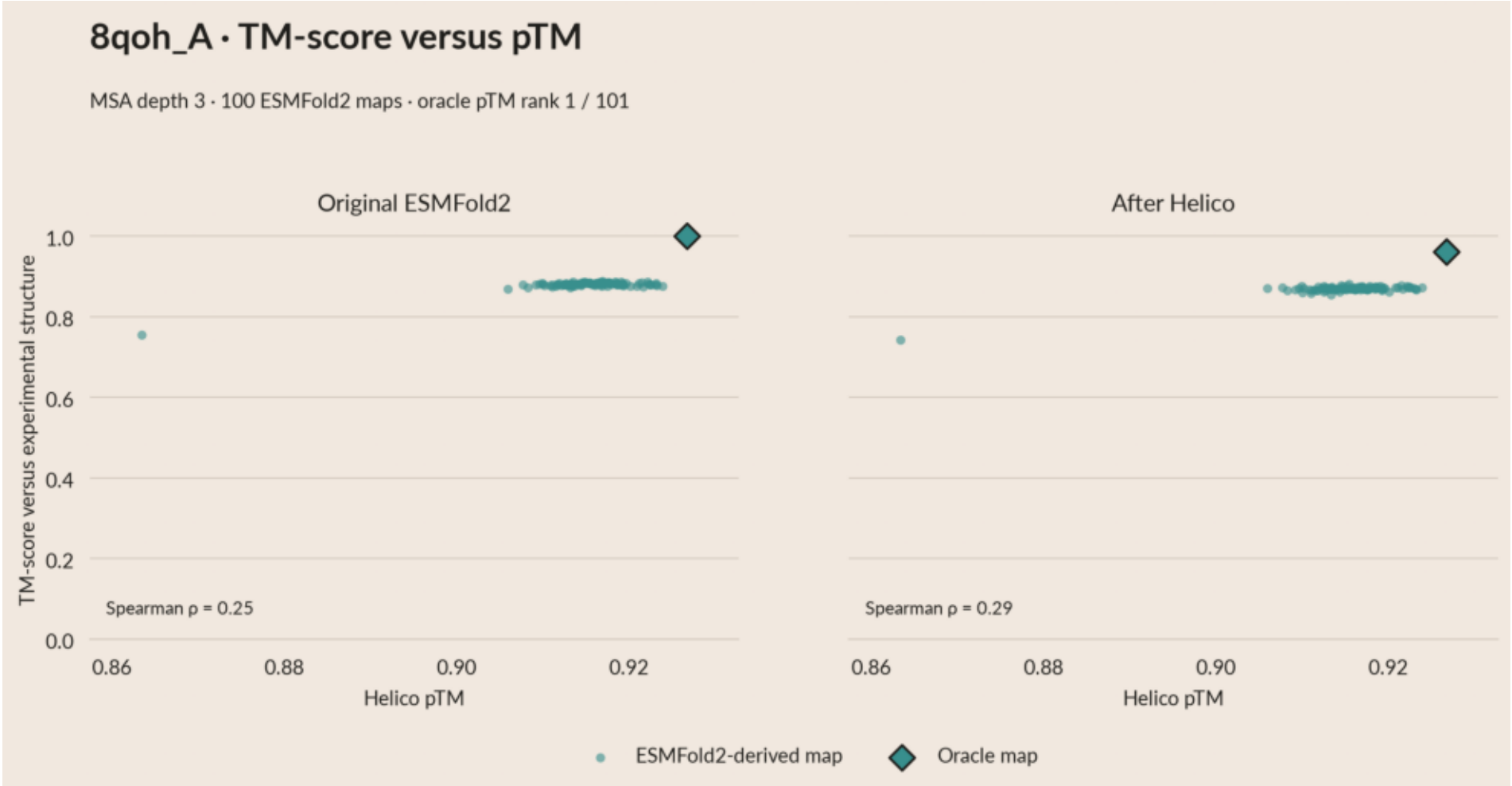
## 02c\_accuracy\_confidence\_protein\_8oxk\_A.png

Highest pTM selects one of three Helico samples for every map; pTM also ranks the 101 maps. Left: original ESMFold2 TM-score (oracle reference = 1). Right: selected Helico reconstruction TM-score. Diamonds identify oracle contacts; correlations use only the 100 ESMFold2 maps. TM-score uses matched protein CA atoms; pTM retains all Helico input tokens. No ipTM or clash penalty.



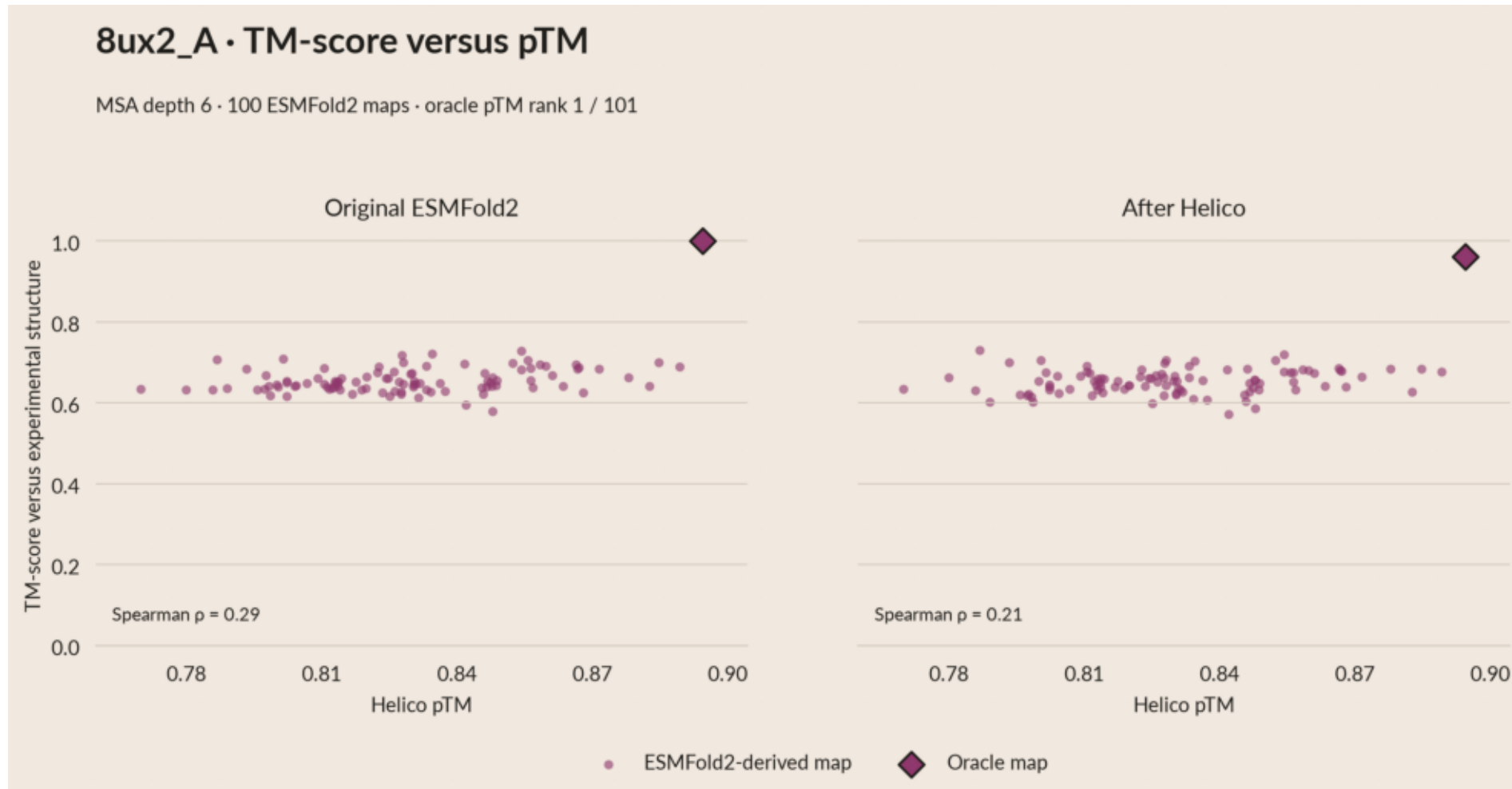
# 02c\_accuracy\_confidence\_protein\_8qoh\_A.png

Highest pTM selects one of three Helico samples for every map; pTM also ranks the 101 maps. Left: original ESMFold2 TM-score (oracle reference = 1). Right: selected Helico reconstruction TM-score. Diamonds identify oracle contacts; correlations use only the 100 ESMFold2 maps. TM-score uses matched protein CA atoms; pTM retains all Helico input tokens. No ipTM or clash penalty.



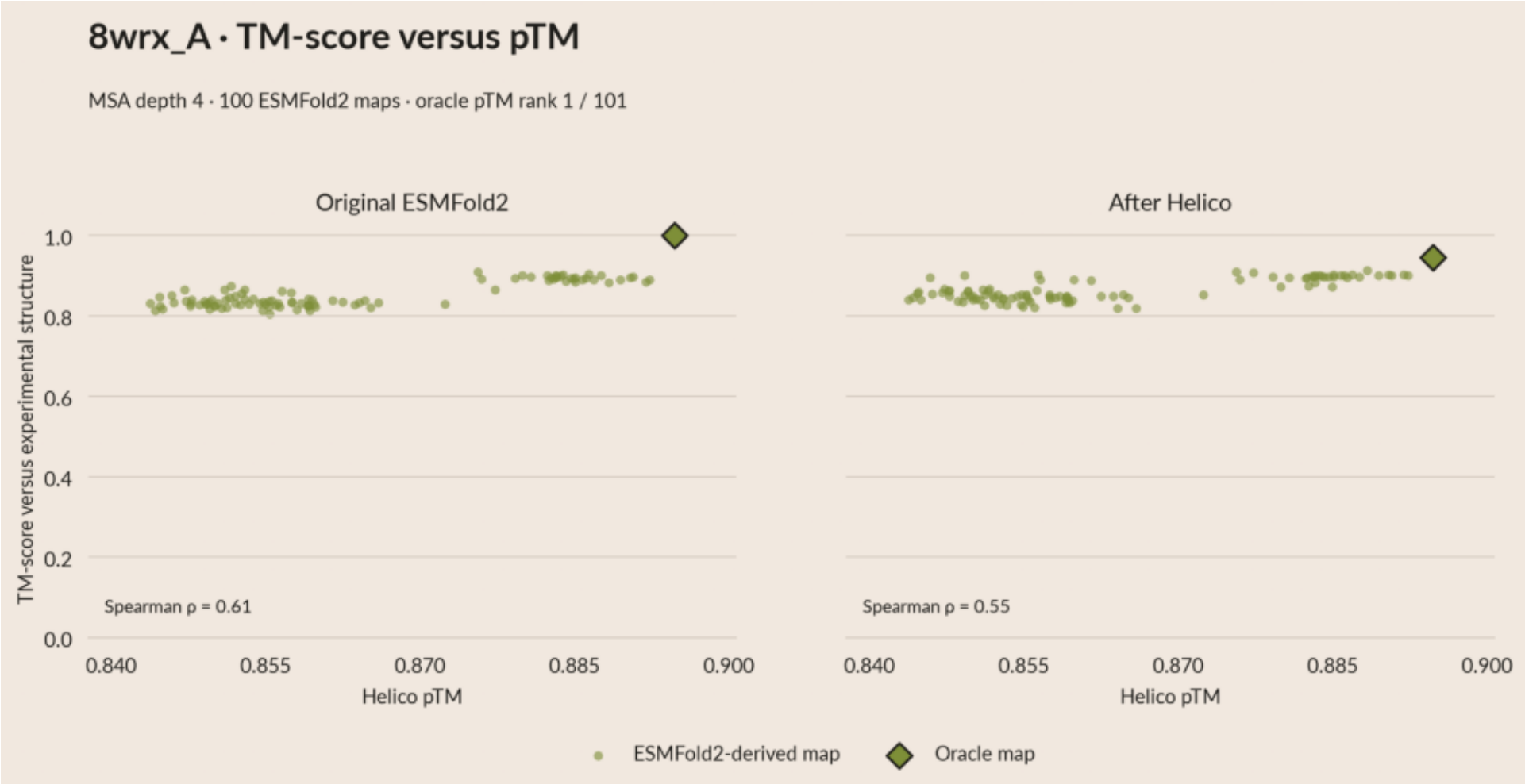
## 02c\_accuracy\_confidence\_protein\_8ux2\_A.png

Highest pTM selects one of three Helico samples for every map; pTM also ranks the 101 maps. Left: original ESMFold2 TM-score (oracle reference = 1). Right: selected Helico reconstruction TM-score. Diamonds identify oracle contacts; correlations use only the 100 ESMFold2 maps. TM-score uses matched protein CA atoms; pTM retains all Helico input tokens. No ipTM or clash penalty.



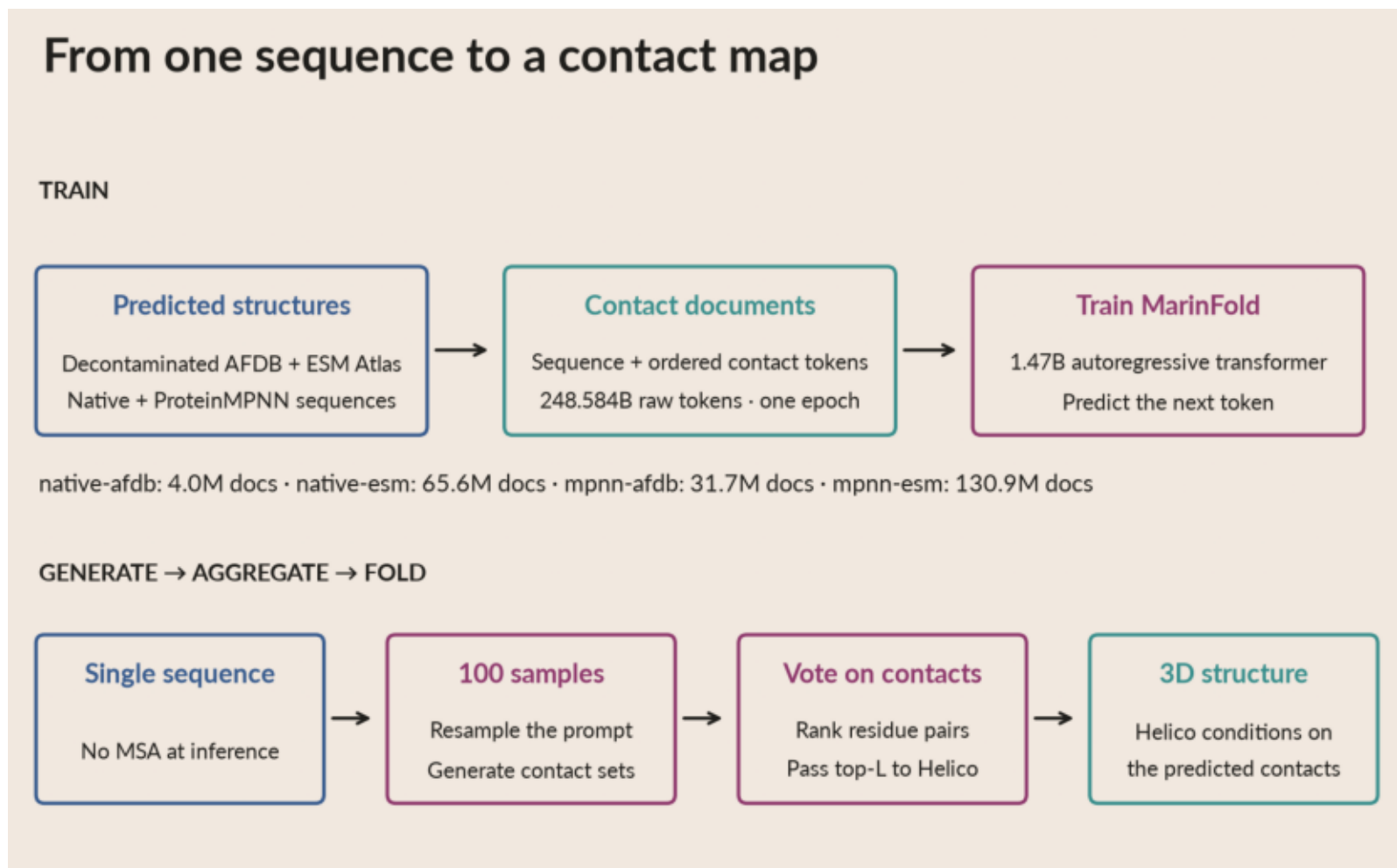
# 02c\_accuracy\_confidence\_protein\_8wrx\_A.png

Highest pTM selects one of three Helico samples for every map; pTM also ranks the 101 maps. Left: original ESMFold2 TM-score (oracle reference = 1). Right: selected Helico reconstruction TM-score. Diamonds identify oracle contacts; correlations use only the 100 ESMFold2 maps. TM-score uses matched protein CA atoms; pTM retains all Helico input tokens. No ipTM or clash penalty.



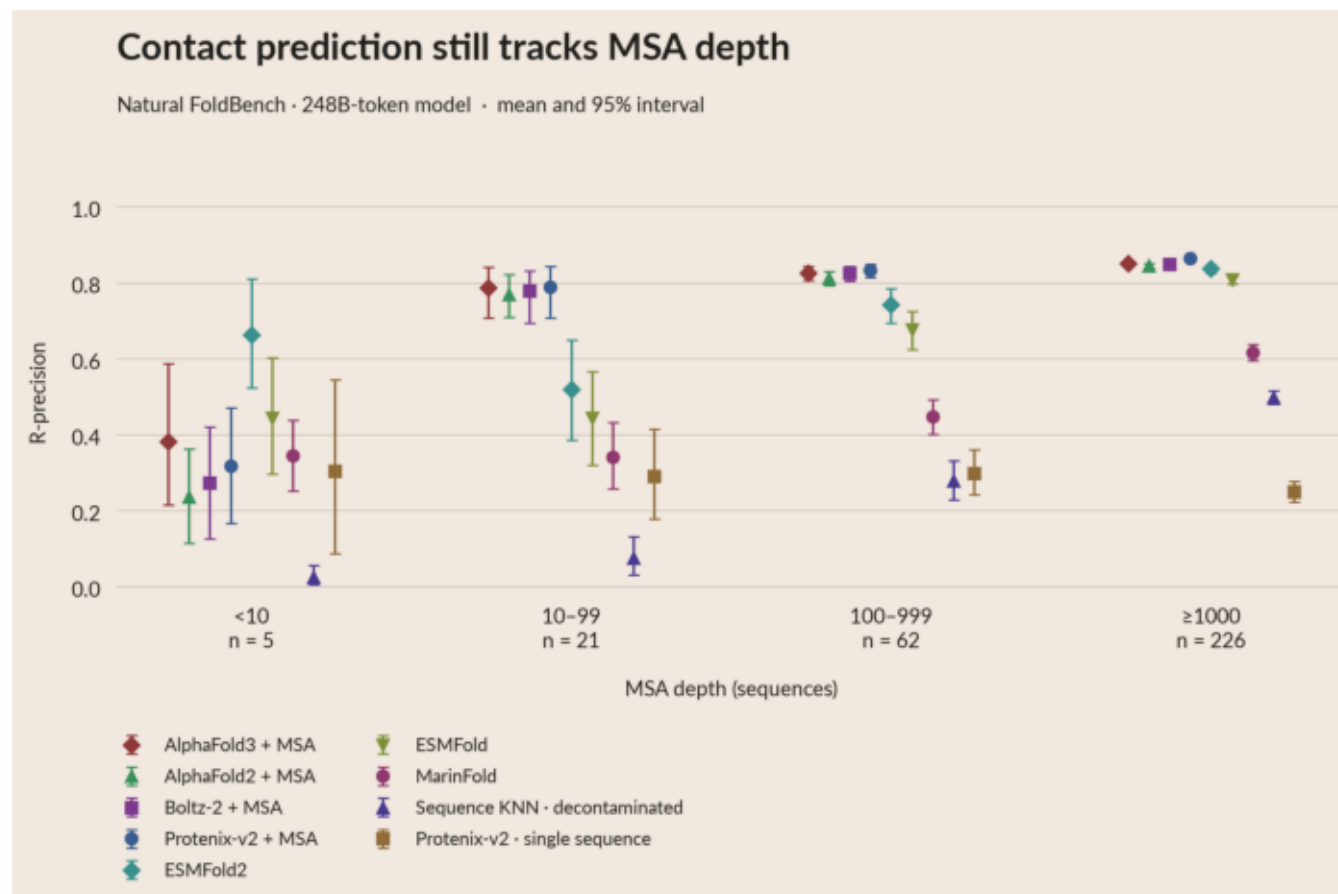
## 03\_method.png

Schematic; arrow widths do not encode amounts. Exp277 step 266,344: a scratch-trained 1.47B-parameter Qwen3, one epoch over 232,090,905 contact documents, totaling 248,583,762,834 raw tokens. Native and ProteinMPNN-redesigned sequences use AFDB / ESM Atlas source structures from the decontaminated corpus. Raw tokens differ from padded training slots.



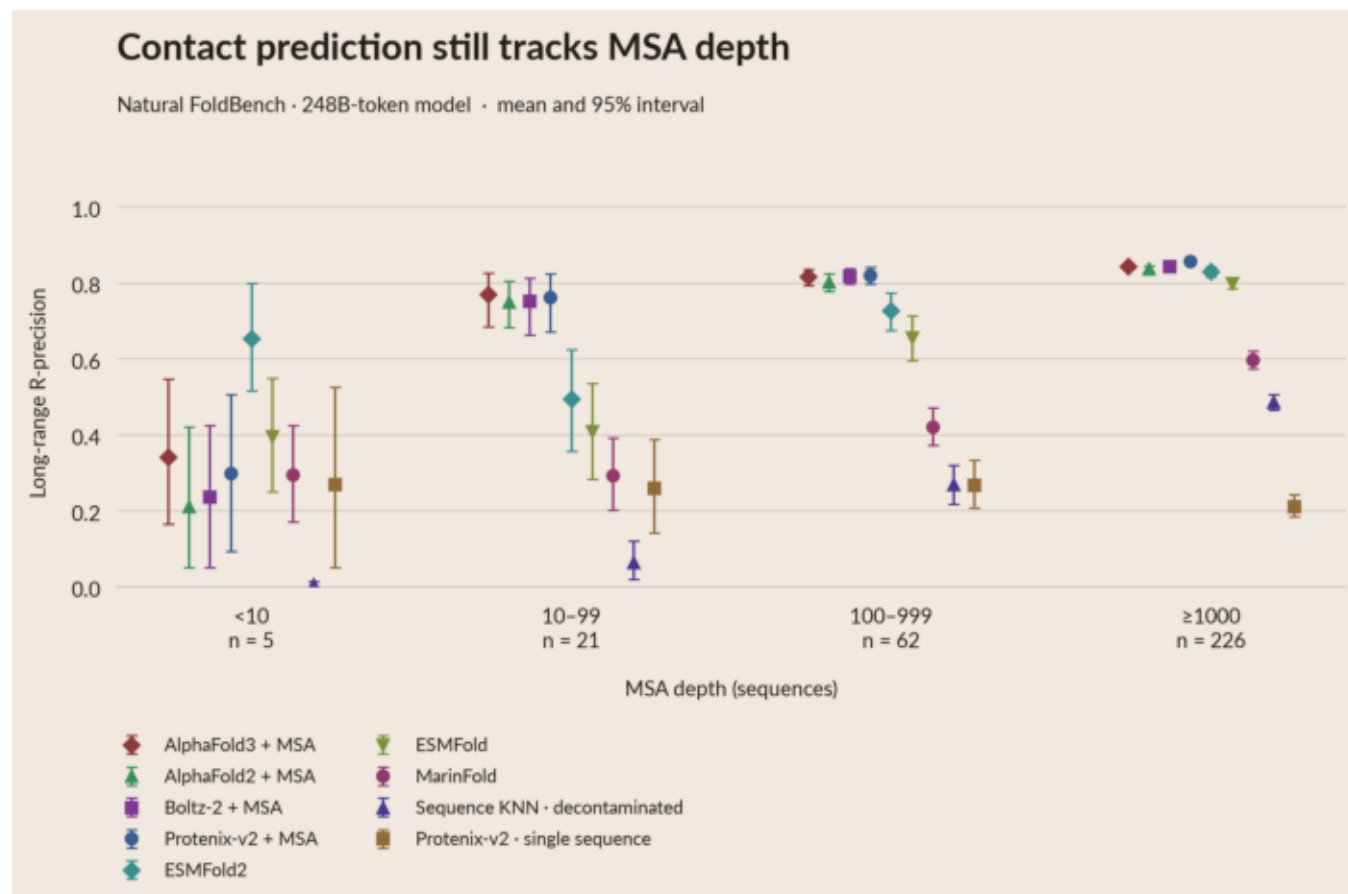
## 04\_contacts.png

Exp277 step 266,344, the 248.584B-raw-token model. All-range R-precision on 314 natural monomers (97 validation + 217 test); 100 resampled rollouts and vote ranking. R is the true-contact count in the resolved candidate universe. KNN indexes the native decontaminated corpus, not the additional redesign sequences. Only five natural proteins have depth <10. The menu also exposes long-range R-precision and separate split/design views. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



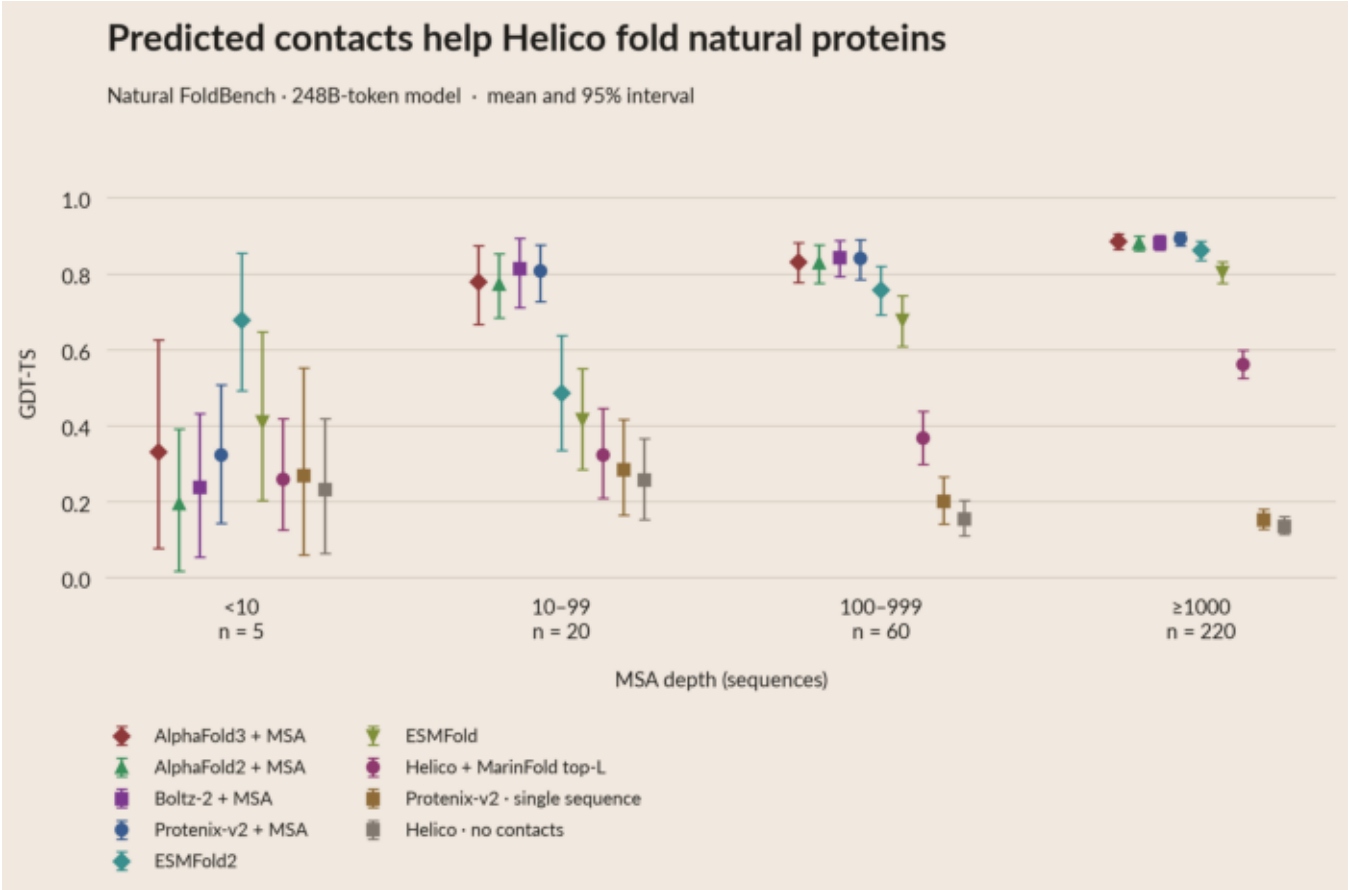
## 04\_contacts\_long.png

Exp277 step 266,344, the 248.584B-raw-token model. All-range R-precision on 314 natural monomers (97 validation + 217 test); 100 resampled rollouts and vote ranking. R is the true-contact count in the resolved candidate universe. KNN indexes the native decontaminated corpus, not the additional redesign sequences. Only five natural proteins have depth <10. The menu also exposes long-range R-precision and separate split/design views. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



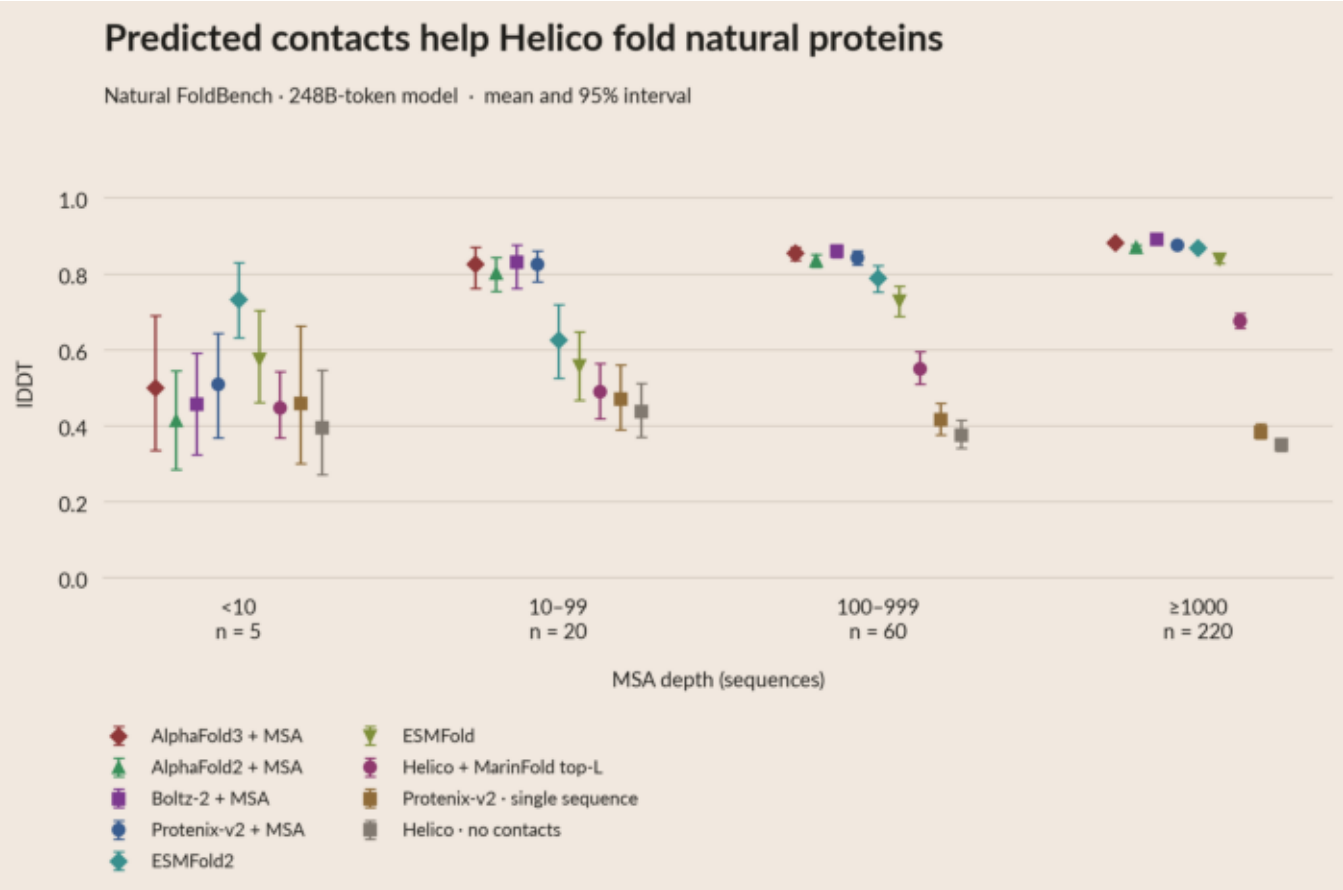
# 05\_folding.png

The 248B-token model through Helico: request the top-L predicted contacts, generate three diffusion samples, select by highest Helico confidence at that fixed cut; no MSA. Helico drops pairs closer than six residues after coordinate mapping; requested and effective counts are saved. Same 305 natural proteins in every arm (95 validation + 210 test). GDT-TS and IDDT are separate metrics. See coverage.csv for exclusions. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



# 05\_folding\_iddt.png

The 248B-token model through Helico: request the top-L predicted contacts, generate three diffusion samples, select by highest Helico confidence at that fixed cut; no MSA. Helico drops pairs closer than six residues after coordinate mapping; requested and effective counts are saved. Same 305 natural proteins in every arm (95 validation + 210 test). GDT-TS and IDDT are separate metrics. See coverage.csv for exclusions. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



## 06\_sampling.png

Exp277 step 266,344, 314 natural proteins (97 validation + 217 test), 100 iid rollouts per protein. Consensus and oracle use the same sample pool within each protein. Individual maps are ranked by emission order; short maps retain denominator R, and malformed or unfinished individual maps score zero. The diagnostic consensus uses all parsed maps, following exp321. Oracle selection uses ground truth. This measures contact accuracy, not distinct folds. Scatter colors identify MSA tiers in the interactive view.

